Circular motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate inward net force to a change in velocity direction.

SUCCESS CRITERIA - I can...

1. Draw tangent velocity and inward acceleration.
2. Use v^2/r or a supplied acceleration.
3. Identify the real interaction providing centripetal force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “centripetal” and “radius” to explain your model or result. Sentence starters are optional.

centripetal: directed toward the center of a circular path

radius: distance from the center to the edge of a circle or sphere

net force: the vector sum of all forces on one object

01 Bring back an earlier idea

A projectile’s velocity is (9, -12) m/s.
Find its speed. What information would locate it?

Word help: projectile = an object moving after launch under the stated forces

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a_c = v^2/r$; $F_{net} = mv^2/r$.
c means centripetal: toward the center. This is the net inward force, not an extra force.

Worked reasoning: A 2 kg object moves at 3 m/s in a circle of radius 3 m. Inward acceleration is $v^2/r = 3\ m/s^2$; net force is 6 N inward.

Your turn: Draw tangent velocity and inward net force at two different points.

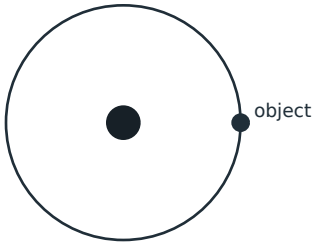
Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Add velocity and acceleration arrows to the marked point on the circle.
Name one interaction that could provide the inward force.

Sentence starter: The ... in my model represents ... . Its label shows ... .



The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Solve and justify

A .50 kg object moves at 4 m/s in a circle of radius 2 m.
Find a_c and net force. Then double speed at same radius.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A string breaks while a ball moves in a circle. Neglect other forces over the next instant.
Sketch its initial path after release.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: 1 2 3

06 Choose a method

A toy model predicts position (-80, 100) km with bounds ±8 km in x and ±10 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

07 Exit check

At fixed speed and mass, double circle radius. How do acceleration and net force change?
Explain from equations.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

The maximum horizontal force on a 0.5 kg object is 8 N. For a 1 m circle, find the maximum speed. Then double the radius and find the new limit.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3